
AGS: CAPSTONE PROJECT

BANKING CUSTOMER SUPPORT AI ASSISTANT

Final Report

Applied GenAI Specialization (AGS)

Submitted By: Maheep Singh

Batch : Gen AI Specialization 2026 cohort 73



Delivered by
simplilearn

Table of Contents

List of Figures	4
List of Tables	4
Section 1: Executive Summary	5
Introduction.....	5
Business Context.....	5
Problem Statement.....	5
Project Deliverables/Tasks.....	5
Solution Overview.....	6
Section 2: System Architecture.....	7
Introduction.....	7
Architecture Diagram.....	7
Functional Component Overview	8
1. User Interaction Layer	8
2. Agent Workflow Layer – LangGraph	9
3. Data & Persistence Layer.....	9
4. State Management & Memory Layer.....	10
5. Monitoring & Analytics Layer	10
LangGraph Workflow Explanation	11
1. Query Classification.....	11
2. Intent Routing	12
3. Specialized Agent Execution	12
4. Customer Intelligence Generation.....	13
5. Response Generation	13
6. Response Evaluation – LLM Judge Agent	14
7. Reflection-Based Improvement	14
8. Final Response Delivery.....	14
Section 3: State Management & Persistence	15

State Management & Memory Layer.....	15
BankingSupportState	15
Streamlit Session State.....	16
LangGraph Memory Persistence	16
Conversation Context	16
Data & Persistence Layer.....	16
Table 1 – Support Tickets	17
Table 2 – CSAT Feedback.....	17
Table 3 – Evaluation Results.....	17
Table 4 – Intent Classification Test Results	17
Section 4: Evaluation, Analytics & Monitoring.....	18
Evaluation.....	18
LLM-as-Judge Evaluation.....	18
Intent Classification Test Case Evaluation.....	19
Analytics.....	19
Monitoring	20
Section 5 – Streamlit User Interface	21
User Interface Overview	21
Tab 1 – Customer Assistant	21
Tab 2 – Response Evaluation Dashboard.....	22
Tab 3 – Customer Profile Dashboard	23
Tab 4 – Agent Execution Trace Dashboard	23
Tab 5 – Intent Classification Evaluation (Run Test-cases)	24
Section 6 – Sample Output Screenshots	25
Intent – Positive.....	25
Intent – Inquiry.....	26
Intent – Negative	27
Section 7 – Key Capabilities & Innovation Highlights	29
Key Capabilities.....	29
Innovation Highlights	30
Section 8 – Challenges Faced & Future Enhancements	31
Challenges Faced	31
Future Enhancements.....	31

List of Figures

Figure 1: Banking Customer Support Assistant Architecture Diagram.....	7
Figure 2: Banking Customer Support Assistant LangGraph Orchestration Workflow	8
Figure 3: Tab 1 – Customer Assistant (before & after new session) Streamlit.....	21
Figure 4: Tab 2 – Response Evaluation Dashboard Streamlit.....	22
Figure 5: Tab 3 – Customer Profile Dashboard Streamlit	23
Figure 6: Tab 4 – Agent Execution Trace Dashboard Streamlit	23
Figure 7: Tab 5 – Intent Classification Evaluation (Run Test-cases) Streamlit	24
Figure 8: Output Screenshot Intent - Positive Casual	25
Figure 9: Output Screenshot Intent - Positive Resolution Confirmation	25
Figure 10: Output Screenshot Intent – Inquiry Ticket ID in Field	26
Figure 11: Output Screenshot Intent – Inquiry No Ticket ID Provided	26
Figure 12: Output Screenshot Intent – Inquiry Ticket ID in User Query	27
Figure 13: Output Screenshot Intent – Negative Complaint for a New Ticket	27
Figure 14: Output Screenshot Intent – Negative Complaint for an Existing Resolved Ticket	28
Figure 15: Output Screenshot Intent – Negative Complaint for an Existing Open Ticket	28

List of Tables

No table of figures entries found.

Section 1: Executive Summary

Introduction

Customer support is a critical function in the banking industry, directly influencing customer satisfaction, trust and retention. Customers expect timely issue resolution, accurate information, personalized interactions and transparent communication throughout the support lifecycle. Traditional support processes often rely on manual ticket handling and static workflows, which can lead to inconsistent customer experiences, delayed responses and limited personalization.

Advancements in Generative AI and Agentic Frameworks provide opportunities to transform customer support operations by enabling intelligent decision-making, contextual understanding, automated workflow execution and continuous response improvement. This project demonstrates how a multi-agent AI system can be leveraged to deliver more efficient, accurate and customer-centric support experiences within a banking environment.

Business Context

Banking support teams handle a wide range of customer interactions, including service requests, incident reporting, status inquiries, complaint handling, and feedback management. These interactions often require support agents to interpret customer intent, validate ticket information, access historical context, determine appropriate actions, and communicate outcomes clearly and empathetically.

Several operational challenges commonly impact support effectiveness:

- Inconsistent interpretation of customer requests.
- Delays in identifying the correct support workflow.
- Limited personalization in customer interactions.
- Lack of awareness of historical customer experiences.
- Difficulty maintaining response quality and consistency.
- Manual effort required to investigate ticket history and customer context.

These challenges can negatively affect customer satisfaction, increase handling times, and create inconsistent service experiences.

Problem Statement

The objective of this project is to design and implement an intelligent Banking Customer Support AI Assistant capable of understanding customer intent, executing appropriate support workflows, generating accurate customer responses, and continuously improving response quality through automated evaluation.

The solution must:

- Correctly identify customer intent and interaction type.
- Support positive feedback, negative feedback, and inquiry scenarios.
- Manage support ticket workflows accurately.
- Generate customer-facing responses grounded in ticket context.
- Incorporate historical customer intelligence to personalize interactions.
- Evaluate response quality using an AI-based judging mechanism.
- Improve responses through iterative self-reflection when required.

The goal is to create a scalable and intelligent customer support system that enhances both operational efficiency and customer experience.

Project Deliverables/Tasks

- Design and implement a multi-agent Banking Customer Support AI Assistant using LangGraph as the orchestration framework.
- Develop an Intelligent Query Classification Agent capable of identifying customer intent, interaction type, ticket references and resolution confirmations from natural language requests.
- Implement specialized workflow agents to handle positive feedback, customer inquiries and negative feedback scenarios through dedicated business workflows.
- Automate support ticket lifecycle actions, including ticket creation, status inquiries, resolution handling and follow-up management.
- Build a Customer Intelligence Framework capable of generating customer-specific insights using historical feedback, interaction patterns and evaluation outcomes.
- Develop a Response Generation Agent that produces context-aware, customer-facing responses using workflow outputs, ticket information and customer intelligence.
- Implement an LLM-as-Judge evaluation framework to assess response quality across predefined business and customer experience dimensions.

- Incorporate a reflection-based improvement mechanism that automatically refines responses when evaluation quality thresholds are not met.
- Establish workflow state management and conversational memory using LangGraph state persistence and checkpointing mechanisms.
- Design a data persistence layer to store support tickets, customer feedback, evaluation results and classifier validation outcomes.
- Develop monitoring and observability capabilities to capture workflow execution traces, routing decisions, ticket operations and evaluation activities.
- Create analytics capabilities that provide customer profiling insights, response quality trends and operational intelligence.
- Build a Streamlit-based user interface supporting customer interactions, workflow monitoring, customer profiling, evaluation reporting and classifier testing.
- Validate workflow behaviour through predefined intent classification test cases.

Solution Overview

- To address these challenges, a multi-agent Banking Customer Support AI Assistant was developed using Agentic AI principles. The solution combines intent classification, workflow automation, customer intelligence generation, response orchestration and automated quality evaluation within a unified support framework.
- The system analyses customer requests, identifies the appropriate support workflow, retrieves relevant ticket information and generates context-aware responses tailored to the customer's situation. Historical customer interactions, feedback patterns, and evaluation results are analysed to generate customer intelligence that helps personalize future responses and improve communication quality.
- An AI Judge evaluates every generated response against predefined quality criteria, including intent handling, workflow compliance, customer experience and factual accuracy. Responses that fail to meet quality expectations are automatically refined through a reflection mechanism before being delivered to the customer.
- The solution demonstrates how Agentic AI can be applied to customer support operations to improve consistency, personalization, accuracy and overall service quality while reducing manual effort and enhancing customer satisfaction.

Section 2: System Architecture

Introduction

- The Banking Customer Support AI Assistant is designed using a multi-agent architecture built on LangGraph. The solution combines intent classification, workflow orchestration, customer intelligence generation, response creation, automated evaluation and persistent data storage to deliver accurate and context-aware customer support interactions.
- The architecture follows a layered design approach consisting of four primary layers:
 1. User Interaction Layer
 2. Agentic Workflow Layer (LangGraph)
 3. Data & Persistence Layer
 4. State Management & Memory
 5. Monitoring & Analytics
- When a customer submits a request through the Streamlit interface, the request is processed by a LangGraph workflow that identifies customer intent, executes the appropriate business workflow, generates customer intelligence, and creates a personalized response. The response is then evaluated by an LLM-as-Judge component to ensure quality, accuracy, and workflow compliance before being presented to the customer.
- The architecture also incorporates persistent storage for support tickets, customer feedback, evaluation results, and session data, enabling historical analysis and continuous improvement of customer interactions.

Architecture Diagram

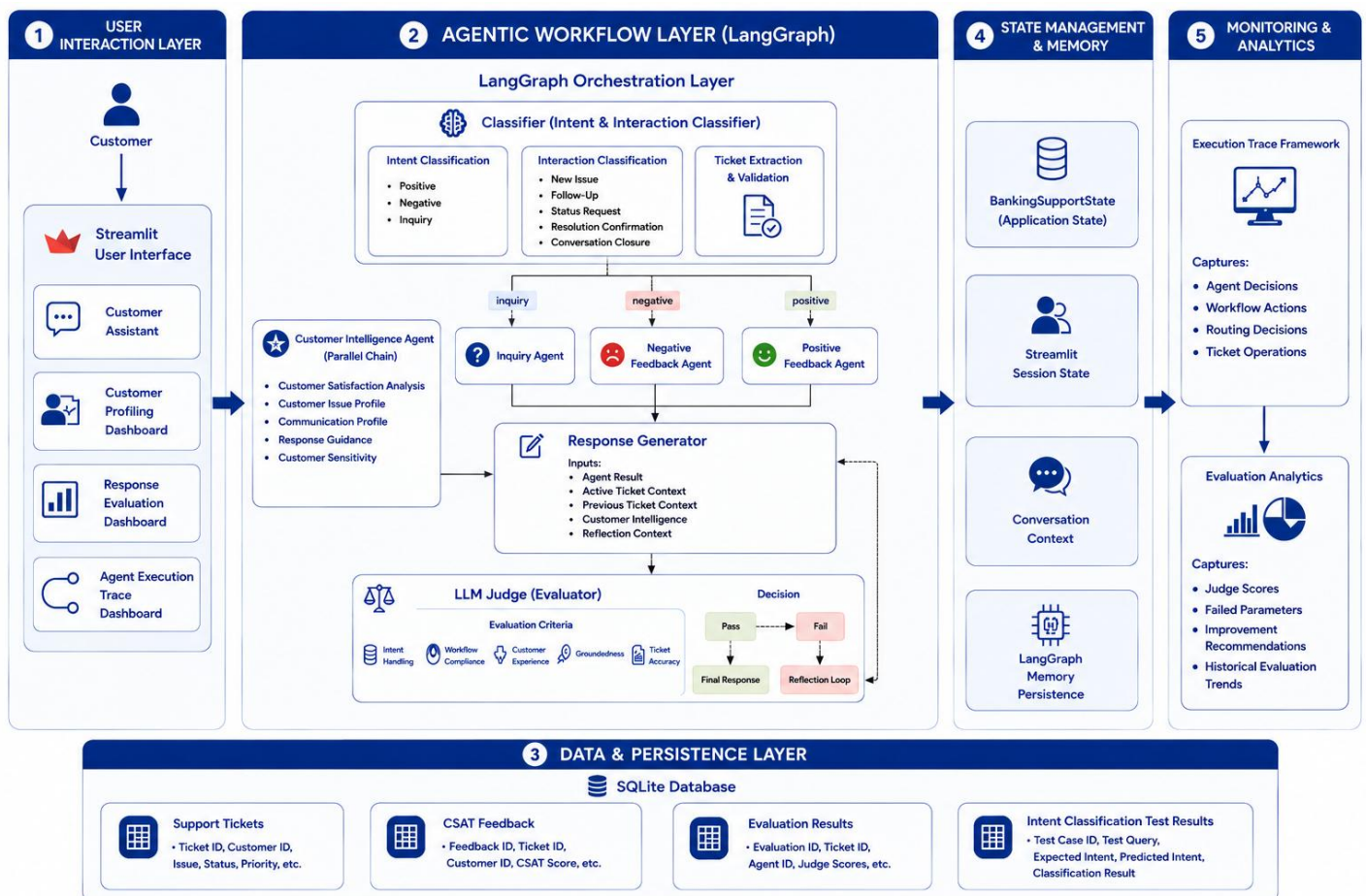


Figure 1: Banking Customer Support Assistant | Architecture Diagram

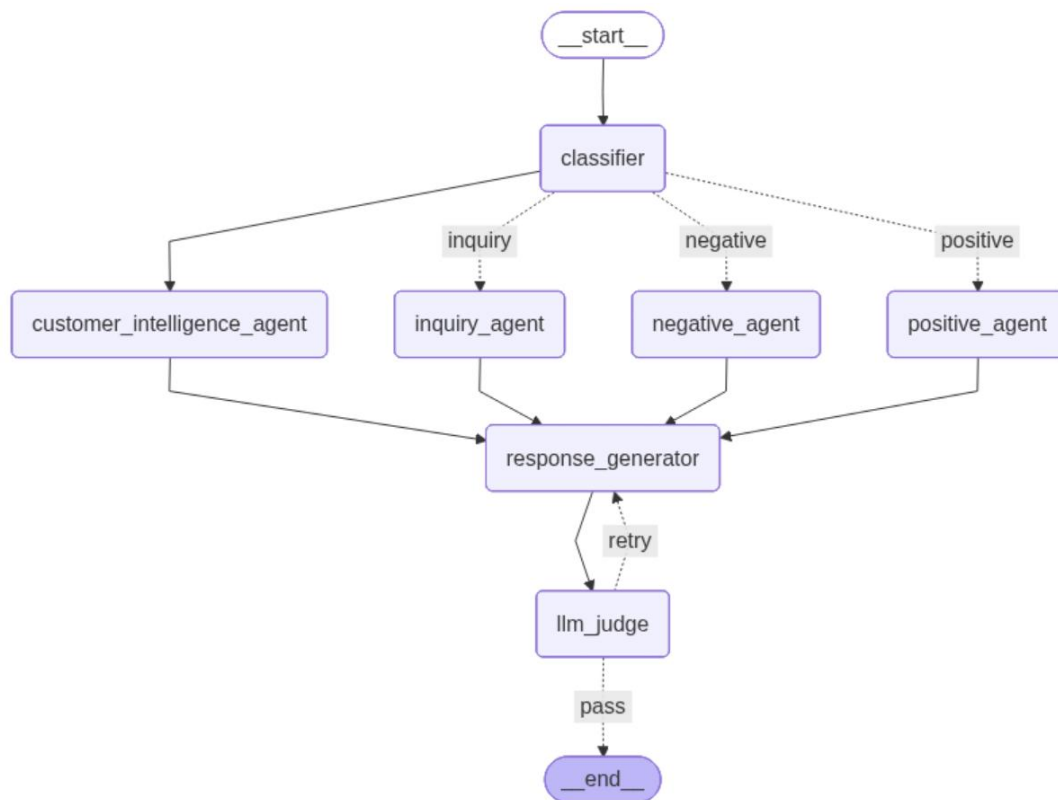


Figure 2: Banking Customer Support Assistant | LangGraph Orchestration Workflow

Functional Component Overview

The Banking Customer Support AI Assistant follows a layered architecture that separates user interaction, workflow orchestration, data persistence, state management, and monitoring capabilities. This separation of responsibilities improves maintainability, scalability, observability, and overall system design.

The solution consists of five major architectural layers: -

1. User Interaction Layer

- **Purpose**
 - Provide a user-friendly interface for customer interactions and operational monitoring.
- **Key Responsibilities**
 - Capture customer details and support requests.
 - Display AI-generated responses.
 - Collect customer feedback.
 - Present customer profiling insights.
 - Display evaluation results and execution traces.
- **Components**
 1. **Customer Assistant** – Primary interface used by customers to submit support tickets and receive AI-generated responses.
 2. **Response Evaluation Dashboard** – Provides visibility into LLM Judge evaluation scores and response quality metrics.
 3. **Customer Profiling Dashboard** – Displays customer intelligence insights derived from historical interactions and feedback.
 4. **Agent Execution Trace Dashboard** – Displays end-to-end workflow execution details for monitoring and troubleshooting.
 5. **Intent Classification Evaluation Dashboard** – Enables systematic testing (run manual test cases) and validation of intent classification accuracy before deployment and during model improvements. Displays expected versus predicted classification results to assess classifier accuracy and consistency.
- **Business Value**
 - Simplifies customer interaction.
 - Improves transparency of AI decisions.
 - Provides visibility into workflow execution and response quality.

2. Agent Workflow Layer – LangGraph

- **Purpose**
 - Act as the central intelligence and orchestration engine of the solution.
- **Key Responsibilities**
 - Classify customer intent and interaction type.
 - Route requests to appropriate workflow agents.
 - Execute ticket management workflows.
 - Generate customer intelligence insights.
 - Generate customer-facing responses.
 - Evaluate response quality.
 - Perform automated response refinement when required.
- **Components**
 1. **Query Classification Agent** – Identifies customer intent, interaction type and ticket references from incoming requests.
 2. **Positive Feedback Agent** – Handles appreciation, positive feedback and resolution confirmation interactions.
 3. **Negative Feedback Agent** – Processes complaints, unresolved issues and customer dissatisfaction scenarios.
 4. **Inquiry Agent** – Handles ticket status requests, information inquiries and general support questions.
 5. **Customer Intelligence Agent** – Analyses historical customer behaviour, satisfaction patterns and communication preferences.
 6. **Response Generator** – Creates the final customer-facing response using workflow outputs and contextual information.
 7. **LLM Judge Agent** – Evaluates response quality against predefined business and customer experience criteria.
 8. **Reflection Loop** – Refines responses using judge feedback until quality requirements are satisfied.
- **Business Value**
 - Automates customer support workflows.
 - Ensures consistent decision-making.
 - Improves response quality through self-evaluation.

3. Data & Persistence Layer

- **Purpose**
 - Store operational, customer and evaluation data required by the workflow in a database (SQLite)
- **Key Responsibilities**
 - Maintain support ticket records.
 - Store customer feedback and CSAT data.
 - Persist evaluation results.
 - Maintain historical customer interaction data.
- **Components (SQLite Database)**
 1. **Support Tickets** – Table storing incident records, ticket status, ownership and resolution details.
 2. **CSAT Feedback** – Table storing customer ratings, comments and satisfaction feedback collected after interactions.
 3. **Evaluation Results** – Table storing LLM Judge scores, evaluation outcomes, and response improvement recommendations.
 4. **Intent Classification Test Results** – Stores predefined test cases, expected classifications, predicted classifications and validation outcomes used to evaluate classifier accuracy.
- **Business Value**
 - Enables historical analysis.
 - Supports workflow continuity.
 - Provides data for customer intelligence generation.

4. State Management & Memory Layer

- **Purpose**
 - Maintain workflow context and conversational continuity throughout execution
- **Key Responsibilities**
 - Manage workflow state.
 - Maintain customer session information.
 - Preserve conversation history.
 - Share context across workflow agents.
 - Support workflow persistence.
- **Components**
 1. **BankingSupportState** – Central workflow state object that stores and shares information across all LangGraph nodes during execution.
 2. **Streamlit Session State** – Maintains user-specific data and application context throughout an active Streamlit session.
 3. **Conversation Context** – Preserves historical interaction history to enable context-aware responses and follow-up handling.
 4. **LangGraph Memory Persistence** – Provides persistent workflow memory to support stateful execution across multiple interactions and processing steps.
- **Business Value**
 - Ensures context-aware decision making.
 - Supports stateful customer interactions.
 - Improves workflow consistency.

5. Monitoring & Analytics Layer

- **Purpose**
 - Provide visibility into workflow execution and response quality.
- **Key Responsibilities**
 - Track agent execution.
 - Capture workflow decisions.
 - Record routing outcomes.
 - Monitor ticket operations.
 - Store evaluation scores and recommendations.
 - Analyse historical response quality trends.
- **Components**
 1. **Execution Trace Framework** – Captures Agent Decisions, Workflow Actions, Routing Decisions, Ticket Operations
 2. **Evaluation Analytics** – Captures Judge Scores, Failed Parameters, Improvement Recommendations, Historical Evaluation Trends
- **Business Value**
 - Improves auditability.
 - Supports troubleshooting and debugging.
 - Enables continuous improvement of AI-generated responses.

LangGraph Workflow Explanation

- The Banking Customer Support AI Assistant utilizes LangGraph as the orchestration framework to coordinate multiple specialized AI agents and workflow components.
- The workflow follows a structured execution path beginning with customer request analysis and ending with response quality validation.
- Each node within the workflow performs a dedicated responsibility, enabling the solution to separate intent understanding, business workflow execution, customer intelligence generation, response creation and quality assurance.
- The overall workflow consists of the following stages:
 1. Query Classification
 2. Intent Routing
 3. Specialized Agent Execution
 4. Customer Intelligence Generation
 5. Response Generation
 6. Response Evaluation
 7. Reflection-Based Improvement
 8. Final Response Delivery
- The following sections describe the purpose and logic implemented within each workflow component: -

1. Query Classification

- **Purpose**
 - ✓ The Query Classification Agent acts as the entry point of the workflow and is responsible for understanding the customer's request before any business processing occurs.
- **Inputs**
 - ✓ Customer Query
 - ✓ Customer Name
 - ✓ Customer Email
 - ✓ Conversation Context
- **Processing Logic**
 - ✓ The agent analyses the incoming customer message and determines:
 1. **Intent Classification**
 - Identifies the overall sentiment and purpose of the interaction.
 - Supported intents: Positive, Negative, Inquiry
 2. **Interaction Classification**
 - Determines the operational context of the request.
 - Supported interaction types: new issue, follow-up, status-request, resolution confirmation, conversation closure
 3. **Ticket Extraction and Validation**
 - Identifies any ticket references provided by the customer and validates the ticket format before workflow execution continues.
- **Outputs**
 - ✓ **Intent** – Identifies the overall purpose or sentiment of the customer request (Positive, Negative, or Inquiry).
 - ✓ **Interaction Type** – Determines the operational context of the request, such as a new issue, follow-up
 - ✓ **Resolution Confirmation Flag** – Indicates whether the customer has explicitly confirmed that their issue has been resolved.
 - ✓ **Ticket ID** – Captures any support ticket reference identified within the customer query.
 - ✓ **Ticket Validation Status** – Confirms whether the extracted ticket reference follows the expected ticket format and validation rules.
 - ✓ **Classification Reasoning** – Provides the rationale used by the classifier to support the intent and interaction type determination.
- **Business Value**
 - ✓ Provides accurate routing decisions and ensures the correct workflow is executed for each customer request.

2. Intent Routing

- **Purpose**
 - ✓ The Intent Router directs requests to the appropriate business workflow based on the classification results generated by the Query Classification Agent.
- **Processing Logic**
 - ✓ The router evaluates the identified intent and selects the corresponding workflow path.
 - ✓ Routing Rules: -
 - Positive → Positive Feedback Agent
 - Negative → Negative Feedback Agent
 - Inquiry → Inquiry Agent
- **Outputs**
 - ✓ Selected Workflow Path
 - ✓ Assigned Agent
- **Business Value**
 - ✓ Ensures that each customer request is processed by the most appropriate specialized agent.

3. Specialized Agent Execution

Inquiry Agent

- **Purpose**
 - ✓ Handles customer requests seeking information regarding support tickets or general service inquiries.
- **Processing Logic**
 - ✓ The agent retrieves ticket information from the database.
 - ✓ Validates ticket ownership and existence.
 - ✓ Retrieves current ticket status.
 - ✓ Identifies the latest available ticket updates.
 - ✓ Determines the appropriate workflow action (i.e., existing_ticket_used or no_ticket_reference))
- **Outputs**
 - ✓ Agent Response regarding Inquiry Response Data
 - ✓ Ticket Context
 - ✓ Workflow Action
- **Business Value**
 - ✓ Provides customers with accurate and timely information regarding their support requests.

Negative Feedback Agent

- **Purpose**
 - ✓ Handles customer complaints, dissatisfaction, unresolved issues and service-related concerns.
- **Processing Logic**
 - ✓ The agent evaluates Current customer concern, Existing ticket status (Resolved or not), Historical ticket context
 - ✓ Based on the analysis, the agent determines whether:
 - An existing ticket should continue to be used (if it is in 'Open' state).
 - A new ticket should be created (if no ticket reference).
 - A follow-on ticket should be created for a previously Resolved issue (as the system doesn't allow to reopen a ticket).
 - ✓ Supported workflow actions:
 - existing_ticket_used
 - new_ticket_created
 - new_ticket_created_from_resolved_ticket
- **Outputs**
 - ✓ Agent Response
 - ✓ Active Ticket Context
 - ✓ Previous Ticket Context (if applicable)
 - ✓ Workflow Action
 - ✓ Resolution Information
- **Business Value**
 - ✓ Ensures unresolved customer concerns are handled consistently and aligned with support operations.

Positive Feedback Agent

- **Purpose**
 - ✓ Handles customer appreciation, positive feedback and issue resolution confirmations.
- **Processing Logic**
 - ✓ The agent evaluates whether the customer is:
 - Expressing appreciation.
 - Confirming issue resolution.
 - Closing an interaction positively.
 - ✓ When resolution is confirmed, the corresponding support ticket is updated to a Resolved state.
- **Outputs**
 - ✓ Agent Response
 - ✓ Resolution Confirmation Status
 - ✓ Ticket Updates
 - ✓ Workflow Action (existing_ticket_resolved)
- **Business Value**
 - ✓ Captures customer satisfaction signals and supports ticket lifecycle management for Resolved tickets

4. Customer Intelligence Generation

- **Purpose**
 - ✓ Generates customer-specific intelligence using historical interaction customer data (feedback records) and LLM evaluation history.
- **Processing Logic**
 - ✓ The agent analyses:
 - Customer satisfaction trends
 - Historical issue patterns
 - Customer feedback comments
 - Communication preferences
 - Previous evaluation results
 - ✓ The analysis is converted into structured customer intelligence that can be consumed by downstream agents.
- **Outputs**
 - ✓ Customer Satisfaction Summary
 - Provides an assessment of the customer's overall satisfaction trend, relationship health and key improvement opportunities based on historical feedback and evaluation patterns.
 - ✓ Customer Issue Profile
 - Identifies recurring issue categories, inquiry patterns and operational concerns frequently raised by the customer.
 - ✓ Customer Communication Profile
 - Summarizes the customer's communication expectations and preferences derived from historical feedback comments.
 - ✓ Customer Response Guidance
 - Provides actionable recommendations to help downstream agents tailor tone, empathy, level of detail and response structure for future interactions.
 - ✓ Customer Sensitivity
 - Indicates the overall relationship risk level (LOW, MEDIUM, or HIGH) and the degree of care required when handling future customer interactions.
- **Business Value**
 - ✓ Enables personalized and context-aware customer interactions.

5. Response Generation

- **Purpose**
 - ✓ Creates the final customer-facing response using upstream agent outputs and customer intelligence.
- **Processing Logic**
 - ✓ The agent combines Customer Query, Agent Results, Active Ticket Context, Previous Ticket Context, Customer Intelligence, Workflow Action, Reflection Feedback
 - ✓ The generated response follows predefined business rules to ensure:
 - Ticket accuracy
 - Workflow compliance
 - Customer-friendly communication
 - Context awareness

- **Outputs**
 - ✓ Draft Customer Response
- **Business Value**
 - ✓ Generates consistent, personalized and policy-compliant customer communications.

6. Response Evaluation – LLM Judge Agent

- **Purpose**
 - ✓ Evaluates the quality of generated responses before they are delivered to customers.
- **Processing Logic**
 - ✓ The judge evaluates responses across 4 dimensions: -
 - Intent Handling – Measures whether the response correctly addresses the customer's intent, request and interaction context.
 - Workflow Compliance – Measures whether the response adherences to predefined ticket management workflows and business process rules.
 - Customer Experience – Measures the quality of customer communication, including clarity, empathy, professionalism, and personalization.
 - Groundedness – Measures whether all information in the response is supported by available system context without hallucinations or fabricated details.
 - ✓ The agent assigns scores and determines whether the response meets the required quality threshold.
 - ✓ The response must achieve a score of 9 (or more) in EACH parameter in order to be considered 'Pass'.
- **Outputs**
 - ✓ Evaluation Scores for each dimension out of 10
 - ✓ Failed Parameters
 - ✓ Improvement Recommendations
 - ✓ Pass / Fail Decision
- **Business Value**
 - ✓ Provides automated quality assurance and reduces the risk of inaccurate customer responses.

7. Reflection-Based Improvement

- **Purpose**
 - ✓ Improves response quality when evaluation requirements are not met.
- **Processing Logic**
 - ✓ When the LLM Judge identifies deficiencies & returns 'Fail', the flow is routed back to the Response Generator for revised output.
 - ✓ Improvement Recommendations & Failed Parameters are fed to response generator as an additional context for further improvement
 - ✓ The Response Generator produces a revised response based on the feedback.
 - ✓ The evaluation cycle continues until the response satisfies the required quality criteria.
 - ✓ To ensure there is no infinite loop in the reflection mechanism, maximum iteration limit is set to 5.
- **Outputs**
 - ✓ Revised Response
 - ✓ Updated Evaluation Results
- **Business Value**
 - ✓ Creates a self-improving workflow capable of producing higher-quality customer responses with minimal manual intervention

8. Final Response Delivery

- **Purpose**
 - ✓ Delivers the approved response to the customer through the Streamlit interface.
- **Processing Logic**
 - ✓ Only responses that satisfy evaluation requirements are presented to the customer.
 - ✓ The interaction outcome, evaluation results and customer feedback are then persisted for future analysis into the database.
- **Outputs**
 - ✓ Final Customer Response
 - ✓ Evaluation Record in the table 'evaluation_results'.
 - ✓ Customer Feedback Record (if submitted from user interface) in the table 'csat'.
- **Business Value**
 - ✓ Ensures customers receive accurate, evaluated and context-aware support responses.

Section 3: State Management & Persistence

- The Banking Customer Support AI Assistant relies on a combination of workflow state management, session persistence, conversational memory and database storage to support intelligent, context-aware customer interactions.
- This layer ensures that information generated during workflow execution is preserved, shared across agents and made available for future processing and analysis.
- The solution incorporates two key architectural components: -
 1. State Management & Memory Layer
 2. Data & Persistence Layer
- Together, these components provide workflow continuity, contextual understanding, operational persistence and historical analysis capabilities.
- The State Management & Persistence Layer serves as the foundation for workflow continuity, contextual awareness, operational storage, and quality monitoring. Together, these capabilities enable the Banking Customer Support AI Assistant to deliver accurate, personalized and stateful customer interactions while maintaining complete traceability across workflow execution.

State Management & Memory Layer

- **Purpose**
 - The State Management & Memory Layer is responsible for maintaining workflow context, preserving customer interactions and enabling information sharing between LangGraph nodes throughout the execution lifecycle.
- **Key Responsibilities**
 - Maintain workflow state across agents.
 - Enable information sharing between workflow nodes.
 - Preserve customer interaction context.
 - Support stateful multi-step workflows.
 - Provide workflow memory persistence.
- **Components**
 1. BankingSupportState – Central workflow state object used to exchange information across all LangGraph nodes.
 2. Streamlit Session State – Maintains user-specific data and application context during an active Streamlit session.
 3. LangGraph Memory Persistence – Provides persistent workflow memory to support stateful execution across interactions.
 4. Conversation Context – Preserves historical interaction history required for context-aware response generation.
- **Business Value**
 - Enables context-aware customer interactions.
 - Reduces information loss between workflow steps.
 - Improves workflow consistency and continuity.
 - Supports personalized customer experiences.

BankingSupportState

- Acts as the central workflow state object and serves as the single source of truth throughout workflow execution. Provides a centralized mechanism for sharing information across workflow agents while ensuring consistency throughout execution.
- Centrally stores the following: -
 - Customer Context – name, email, ticket id (if any), unique thread id
 - Current Context – current user query
 - Ticket Validation – flag to validate ticket id
 - Customer Intent – intent, interaction type, classification reasoning, resolution confirmation flag
 - Ticket Context – ticket status, description, tech. comments, created/updated date, workflow action, previous ticket context
 - Agent Output – Intermediary output from Positive, Negative or Inquiry Agent
 - Final Response – User facing response
 - Execution Traces for monitoring & troubleshooting
 - Evaluations – scores & reasoning across all 4 dimensions, overall score, response strengths & improvements, judge status (pass/fail), failed parameters, reflection counts
 - Customer Intelligence – all 5 metrics (discussed in previous section)

Streamlit Session State

- Maintains user-specific information throughout an active application session.
- Improves usability by maintaining continuity across user interactions and dashboard views.
- Responsible for preserving customer identity, maintaining session continuity, storing workflow outputs for dashboard display & supporting seamless user interactions.
- Key information maintained: -
 - Customer Name
 - Customer Email
 - Session Information
 - Customer Intelligence Results
 - Evaluation Results
 - Current Conversation Data

LangGraph Memory Persistence

- Provides persistent workflow memory and state retention using LangGraph checkpointing mechanisms.
- Improves workflow reliability and supports long-running customer interactions.
- Key Responsibilities: -
 - Store workflow state.
 - Support workflow recovery.
 - Maintain conversational memory.
 - Enable stateful workflow execution.

Conversation Context

- Maintains historical conversation data required for contextual understanding and follow-up handling.
- Allows the solution to generate responses that consider previous interactions and customer context.
- Key Responsibilities: -
 - Preserve conversation history.
 - Support multi-turn interactions.
 - Enable context-aware response generation.
 - Improve customer experience continuity.
- Key Information Maintained
 - User Messages
 - AI Responses
 - Historical Conversation Flow

Data & Persistence Layer

- **Purpose**
 - The Data & Persistence Layer provides long-term storage for operational, customer, evaluation and testing data
- **Key Responsibilities**
 - Store support ticket information.
 - Store customer feedback.
 - Store response evaluation results.
 - Store classifier validation results.
- **Technology**
 - SQLite Database
- **Components**
 - Support Tickets Table – Stores ticket details, ownership, status, issue descriptions and resolution information.
 - CSAT Feedback Table – Stores customer ratings, comments and satisfaction feedback collected after interactions.
 - Evaluation Results Table – Stores LLM Judge scores, failed parameters, evaluation outcomes and improvement recommendations.
 - Intent Classification Test Results Table – Stores predefined classifier test cases and their validation outcomes.
- **Business Value**
 - Enables historical analysis.
 - Supports workflow continuity.
 - Provides operational persistence.
 - Facilitates testing and governance activities.

Table 1 – Support Tickets

- Stores support tickets throughout their lifecycle.
- Provides the operational foundation for ticket management workflows
- Information Stored: -
 - Ticket ID
 - Customer Name
 - Customer Email
 - Ticket Description
 - Ticket Status
 - Customer Comments
 - Technical Comments
 - Created Timestamp
 - Updated Timestamp

Table 2 – CSAT Feedback

- Stores customer satisfaction feedback and interaction outcomes generated during support engagements.
- Supports customer satisfaction analysis, trend identification, and customer intelligence generation.
- Information Stored: -
 - Feedback Rating
 - Feedback Comment
 - Intent Category
 - Workflow Ticket Action
 - Overall Evaluation Score
 - Associated Ticket ID
 - Timestamp

Table 3 – Evaluation Results

- Stores evaluation results generated by the LLM Judge Agent.
- Provides visibility into response quality and supports continuous improvement initiatives
- Information Stored: -
 - Intent Handling Score & Reasoning
 - Workflow Compliance Score & Reasoning
 - Customer Experience Score & Reasoning
 - Groundedness Score & Reasoning
 - Overall Score
 - Failed Parameters
 - Response Strengths
 - Improvement Recommendations
 - Evaluated Query
 - Generated Response
 - Timestamp
 - Thread ID

Table 4 – Intent Classification Test Results

- Stores classifier validation scenarios and testing outcomes used to assess classification accuracy.
- Supports classifier validation, quality assurance and ongoing model performance monitoring.
- Information Stored: -
 - Test Case ID
 - Test Query
 - Expected Intent
 - Predicted Intent
 - Classification Result

Section 4: Evaluation, Analytics & Monitoring

- The Banking Customer Support AI Assistant incorporates a comprehensive Evaluation, Analytics and Monitoring framework to ensure the solution operates reliably, delivers high-quality customer interactions, and provides complete operational transparency.
- The Evaluation capability focuses on validating both response quality and workflow accuracy. Responses generated by the AI Assistant are assessed using an LLM-as-Judge framework, while predefined test scenarios are used to validate intent classification performance and workflow routing behaviour. Together, these mechanisms help ensure that customer requests are correctly understood, routed, and handled in accordance with business requirements.
- The Analytics capability transforms historical interaction, feedback and evaluation data into actionable insights. These insights help identify customer behaviour patterns, recurring service concerns, communication preferences, response quality trends and opportunities for continuous improvement.
- The Monitoring capability provides end-to-end visibility into workflow execution by capturing execution traces across all major workflow stages. This enables users to understand how decisions are made, validate workflow behaviour, troubleshoot issues and maintain operational governance.

Evaluation

- Evaluate the quality, accuracy, and reliability of AI-generated outputs while ensuring customer requests are correctly classified and routed through the appropriate workflow.
- The solution incorporates two complementary evaluation mechanisms: -
 - LLM-as-Judge Evaluation
 - Intent Classification Test Case Evaluation
- Together, these mechanisms assess both response quality and workflow accuracy.

LLM-as-Judge Evaluation

- **Purpose**
 - To assess the quality of customer-facing responses before they are delivered to customers.
- Every response generated by the AI Assistant is automatically evaluated using an LLM-as-Judge framework.
- The judge assesses the response across four quality dimensions: -
 - Intent Handling – Verifies that the response correctly addresses the customer's request and intent.
 - Workflow Compliance – Verifies that the response adheres to ticket management workflows and business process rules.
 - Customer Experience – Evaluates communication quality, including clarity, professionalism, empathy, and personalization.
 - Groundedness – Verifies that all information presented in the response is supported by available system context.
- Based on the evaluation outcome, the judge assigns scores, identifies improvement opportunities and determines whether the response satisfies the required quality threshold.
- **Reflection Loop**
 - ✓ When a response does not meet the required quality standard: -
 - Failed evaluation dimensions are identified.
 - Improvement recommendations are generated.
 - The response is regenerated using the evaluation feedback.
 - The revised response is re-evaluated.
 - ✓ The solution supports a maximum of 5 evaluation-reflection cycles to achieve the desired response quality while preventing infinite processing loops.
- **Evaluation Outcomes**
 - Dimension Scores with their Reasoning
 - Overall Evaluation Score
 - Failed Parameters
 - Response Strengths
 - Improvement Recommendations
 - Pass / Fail Decision
 - Reflection Count
- **Benefits**
 - Improves response quality and consistency.
 - Reduces inaccurate customer communications.
 - Provides automated quality assurance.
 - Enables continuous improvement through self-correction.

Intent Classification Test Case Evaluation

- **Purpose**
 - To validate the accuracy and reliability of the Query Classification Agent.
 - A predefined set of representative customer scenarios is executed through the Query Classification Agent.
 - For each scenario, the predicted classification is compared against the expected classification to validate classifier accuracy and routing behaviour.
 - This evaluation ensures that customer requests are consistently directed to the appropriate workflow path before downstream processing begins.
- **Evaluation Outcomes**
 - Total Test Cases Executed
 - Passed Test Cases
 - Failed Test Cases
 - Classification Accuracy
- **Benefits**
 - Improves classifier reliability.
 - Reduces incorrect workflow routing.
 - Supports testing and validation activities.
 - Increases confidence in workflow execution outcomes.

Analytics

- **Purpose**
 - To generate actionable insights from historical customer interactions, feedback patterns and evaluation outcomes to support personalized customer engagement and continuous service improvement.
 - The solution leverages accumulated operational and customer data to identify trends, recurring concerns, communication preferences and response quality patterns.
- **Key Analytics Capabilities**
 - **Customer Analytics**
 - ✓ Provides customer-specific intelligence derived from historical interactions and feedback.
 - ✓ The analytics framework helps identify: -
 - Customer satisfaction trends.
 - Recurring support concerns.
 - Communication expectations and preferences.
 - Customer relationship sensitivity.
 - Guidance for future interactions.
 - **Operational Analytics**
 - ✓ Provides visibility into the effectiveness and quality of AI-generated customer support interactions.
 - ✓ The analytics framework helps identify: -
 - Response quality trends.
 - Common improvement opportunities.
 - Areas requiring workflow optimization.
 - Overall effectiveness of customer support interactions
- **Benefits**
 - Enables personalized customer experiences.
 - Supports proactive service improvements.
 - Facilitates data-driven decision making.
 - Helps identify recurring customer and operational issues.
 - Improves overall support effectiveness.

Monitoring

- **Purpose**
 - To provide end-to-end visibility into workflow execution, operational behaviour, and AI decision-making activities.
 - The monitoring framework enables users to understand how customer requests are processed, how workflow decisions are made, and how responses are generated throughout the solution lifecycle.
 - The framework combines real-time execution tracing with persistent application logging to support transparency, troubleshooting, and operational governance.
- **Monitoring Capabilities**
 - The framework provides visibility into:
 - ✓ Query classification activities.
 - ✓ Workflow routing decisions.
 - ✓ Ticket management actions.
 - ✓ Customer intelligence generation.
 - ✓ Response generation activities.
 - ✓ Evaluation and reflection outcomes.
 - ✓ Error handling and recovery activities.
 - The monitoring framework constitutes: -
 1. **Real-Time Execution Tracing**
 - Captures workflow events as they occur during execution.
 - Provides visibility into workflow progression from initial customer request through final response generation.
 - Enables users to understand how decisions are made across different workflow stages.
 2. **Application Logging & Trace Persistence**
 - Maintains a persistent record of workflow execution activities.
 - Stores operational logs, trace information, evaluation outcomes, and system events.
 - Supports post-execution analysis, troubleshooting, debugging, and audit activities.
- **Benefits**
 - Improves operational transparency and observability.
 - Supports troubleshooting and root-cause analysis.
 - Simplifies workflow debugging and validation.
 - Provides persistent operational records for support and maintenance activities.
 - Enhances auditability and governance.
 - Increases confidence in AI-driven decision making.
 - Facilitates continuous monitoring and operational support.

Section 5 – Streamlit User Interface

User Interface Overview

- **Purpose**
 - ✓ The Banking Customer Support AI Assistant utilizes a Streamlit-based web interface to provide an interactive and user-friendly experience for both customer interactions and operational monitoring.
- **Key Capabilities**
 - ✓ Submit customer support ticket queries.
 - ✓ Follow-up conversation on tickets using session management.
 - ✓ Receive AI-generated responses.
 - ✓ Provide customer feedback.
 - ✓ View customer intelligence insights.
 - ✓ Monitor workflow execution.
 - ✓ Review response evaluation results.
 - ✓ Validate classifier performance using test scenarios.
- **Interface Components**
 1. Customer Assistant
 2. Response Evaluation Dashboard
 3. Customer Profile Dashboard
 4. Agent Execution Trace Dashboard
 5. Intent Classification Evaluation (Run Test-cases)

Tab 1 – Customer Assistant

– Screenshot

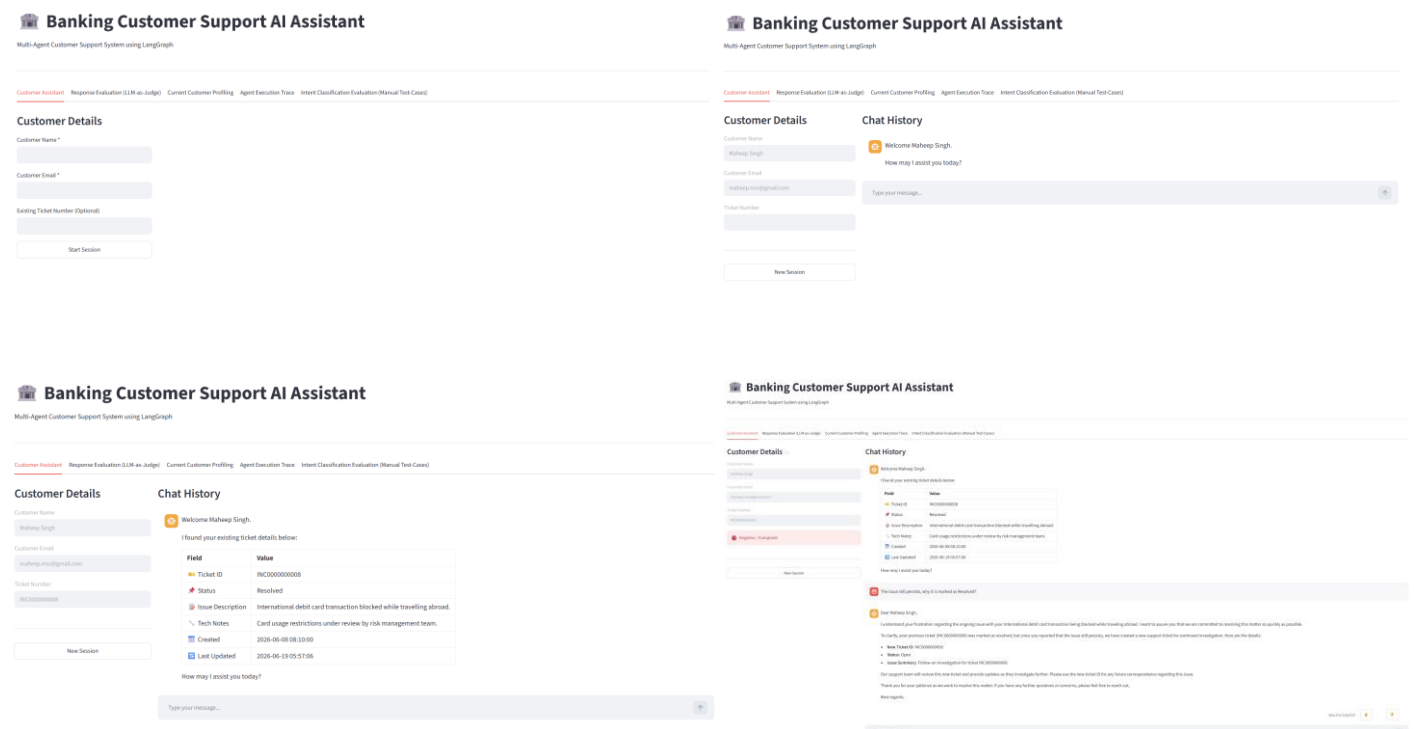


Figure 3: Tab 1 – Customer Assistant (before & after new session) | Streamlit

- **Purpose**
 - Acts as the primary interaction interface between the customer and the AI support system.
- **Key Features**
 - Customer identification using name and email.
 - Conversational chat interface.
 - Support query submission.
 - Ticket inquiry support.
 - Feedback collection.
 - AI-generated customer responses.
 - Display Intent classification at runtime
- **User Benefits**
 - Simple and intuitive support experience.
 - Real-time assistance.
 - Personalized responses based on customer history.

Tab 2 – Response Evaluation Dashboard

- Screenshot

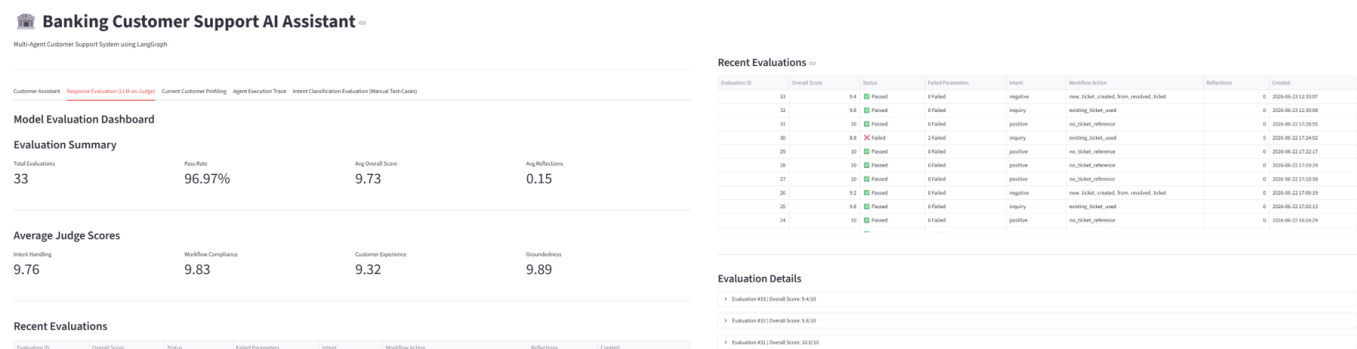


Figure 4: Tab 2 – Response Evaluation Dashboard | Streamlit

- **Purpose**
 - Displays quality assessment results generated by the LLM Judge Agent.
- **Key Information Displayed**
 - Evaluation Summary – Key Metrics
 - Recent Evaluations
 - Each Evaluation Details: -
 - ✓ Conversation Summary
 - ✓ Judge Scores
 - ✓ Judge Reasoning
 - ✓ Failed Parameters
 - ✓ Response Strengths
 - ✓ Response Improvement – Recommendations
- **User Benefits**
 - Provides transparency into response quality.
 - Supports continuous improvement.
 - Enables monitoring of AI performance.

Tab 3 – Customer Profile Dashboard

– Screenshot

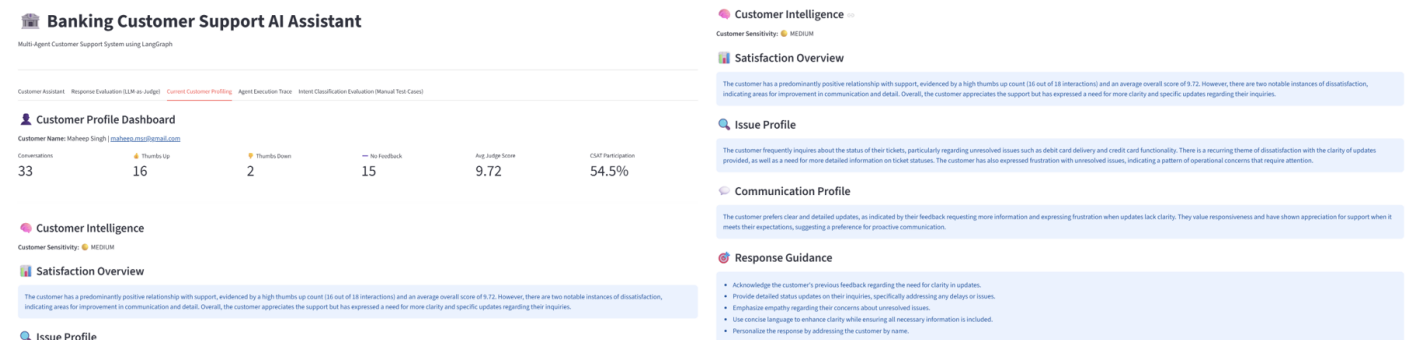


Figure 5: Tab 3 – Customer Profile Dashboard | Streamlit

- **Purpose**
 - Provides customer intelligence generated from historical interactions, feedback patterns and evaluation results.
- **Key Information Displayed**
 - Customer interaction metrics
 - Customer satisfaction analysis
 - Customer issue profile
 - Communication preferences
 - Response guidance
 - Customer sensitivity assessment
- **User Benefits**
 - Enables personalized customer interactions.
 - Provides visibility into customer behaviour and preferences.
 - Supports improved customer experience.

Tab 4 – Agent Execution Trace Dashboard

– Screenshot

Banking Customer Support AI Assistant

Multi-Agent Customer Support System using LangGraph

Customer Assistant Response Evaluation (LLM-as-Judge) Current Customer Profiling **Agent Execution Trace** Intent Classification Evaluation (Manual Test Cases)

Workflow Node Traces

```

===== UI INITIALIZATION: TRACE START =====
LangGraph Instance created
===== UI INITIALIZATION: TRACE END =====

===== UI SESSION: TRACE START =====
Customer Name:Maheep Singh
Customer Email:maheep.ms@gmail.com
Thread ID:maheep_singh_maheep.ms@gmail.com
Session Started
===== UI SESSION: TRACE END =====

===== UI GRAPH: TRACE START =====
Invoking Graph...
===== UI GRAPH: TRACE END =====

===== STARTING CONVERSATION ..... =====
===== CLASSIFIER: TRACE START =====
LangGraph Memory Message Count (0 -> 2 -> 4 ...) = 0
Previous AI Response:None
Intent:positive
Interaction:new
Resolution Confirmation:False
Reasoning:The customer expresses gratitude without confirming any issue resolution.
Ticket ID:None
Ticket Format Valid:None
===== CLASSIFIER: TRACE END =====

===== INTENT ROUTER: TRACE START =====
Classified Intent:positive
Classified Interaction Type:new
Classification Reasoning:The customer expresses gratitude without confirming any issue resolution.
Resolution Confirmation:False
===== INTENT ROUTER: TRACE END =====

===== POSITIVE AGENT: TRACE START =====
Ticket ID = None
No Ticket ID in User query. Generic Thank-you response generated
(Deterministic Response)
===== POSITIVE AGENT: TRACE END =====

===== RESPONSE GENERATOR: TRACE START =====
Intent:positive
Active Ticket ID:None

```

Figure 6: Tab 4 – Agent Execution Trace Dashboard | Streamlit

- **Purpose**
 - Provides complete visibility into workflow execution (Node-level) and agent decision-making.
- **Key Information Displayed**
 - Classification results
 - Routing decisions
 - Workflow actions
 - Ticket operations
 - Customer intelligence generation
 - Response generation steps
 - Judge evaluation outcomes
- **User Benefits**
 - Improves workflow transparency.
 - Supports troubleshooting and debugging.
 - Enables detailed analysis of workflow behaviour.

Tab 5 – Intent Classification Evaluation (Run Test-cases)

- **Screenshot**

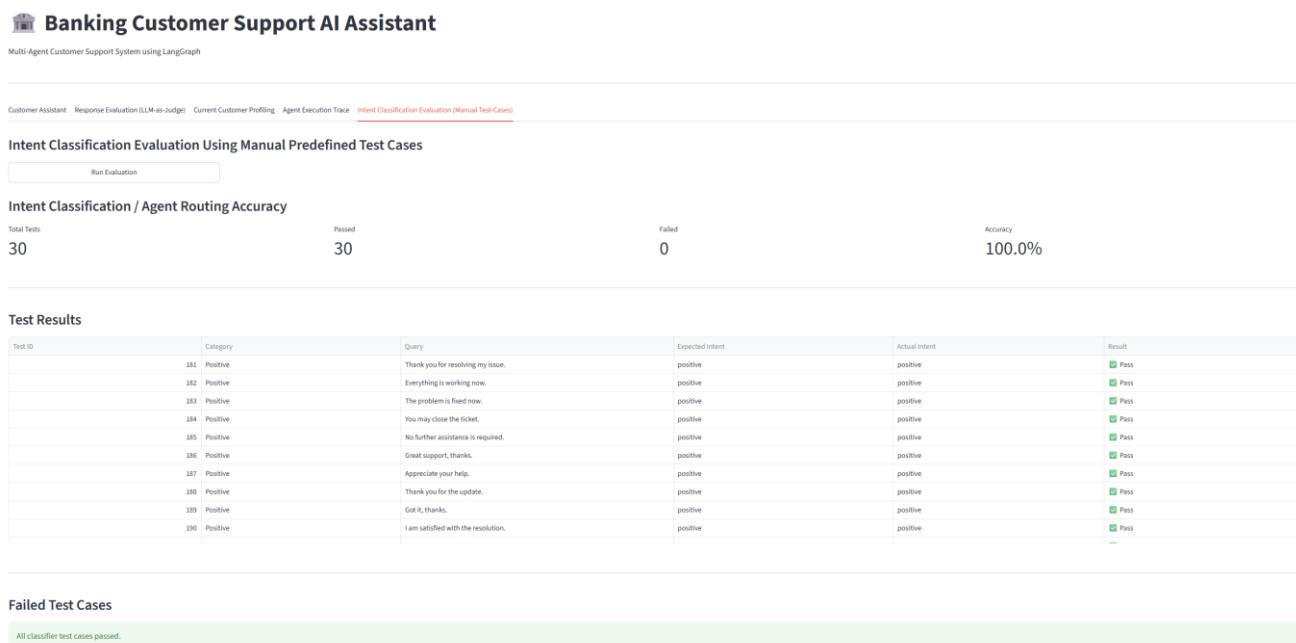


Figure 7: Tab 5 – Intent Classification Evaluation (Run Test-cases) | Streamlit

- **Purpose**
 - Displays detailed classifier evaluation results for all executed test cases
 - Enables automated testing for Intent Classification from predefined test-cases with expected outputs (placed in 'testing' folder)
- **Key Information Displayed**
 - Test Case ID
 - Test Query
 - Expected Intent
 - Predicted Intent
 - Classification Result
 - Pass/Fail Status
- **User Benefits**
 - Provides visibility into classifier performance.
 - Identifies classification failures.
 - Supports automated testing.

Section 6 – Sample Output Screenshots

Intent – Positive

- **Casual Positive feedback without any ticket.**

Banking Customer Support AI Assistant

Multi-Agent Customer Support System using LangGraph

[Customer Assistant](#) [Response Evaluation \(LLM-as-Judge\)](#) [Current Customer Profiling](#) [Agent Execution Trace](#) [Intent Classification Evaluation \(Manual Test-Cases\)](#)

Customer Details

Customer Name

John Smith

Customer Email

john.smith@gmail.com

Ticket Number

Positive / Resolution Confirmation

New Session

Chat History



Welcome John Smith.

How may I assist you today?



Thank you for the support. you all are awesome!



Dear John Smith,

Thank you so much for your kind words! We truly appreciate your positive feedback and are glad to hear that you found our support helpful.

If you have any further questions or need assistance in the future, please don't hesitate to reach out. We're here to help!

Best regards,

Was this helpful?



Type your message...



Figure 8: Output Screenshot | Intent - Positive | Casual

- **Positive feedback followed by confirmation to close the ticket. The ticket was marked as resolved in the system post confirmation from the user**

Banking Customer Support AI Assistant

Multi-Agent Customer Support System using LangGraph

[Customer Assistant](#) [Response Evaluation \(LLM-as-Judge\)](#) [Current Customer Profiling](#) [Agent Execution Trace](#) [Intent Classification Evaluation \(Manual Test-Cases\)](#)

Customer Details

Customer Name

John Smith

Customer Email

john.smith@gmail.com

Ticket Number

INC000000001

Positive / Resolution Confirmation

New Session

Chat History



Welcome John Smith.

I found your existing ticket details below:

Field	Value
📄 Ticket ID	INC000000001
📌 Status	Open
📄 Issue Description	Debit card replacement has not been delivered after 10 business days.
📄 Task Notes	Replacement card dispatch verification pending.
📅 Created	2020-08-01 09:15:00
📅 Last Updated	2020-08-01 09:15:00

How may I assist you today?



I received the debit card today. Thank you for the support.



Dear John Smith,

Thank you for your kind words! We're glad to hear that you received your debit card.

Please note that your ticket remains active as our team is still working on the issue regarding the replacement card dispatch verification. We appreciate your patience and will keep you updated on any progress.

If you have any further questions or need assistance, feel free to reach out.

Best regards,



you may close the ticket as I have resolved the card.



Dear John Smith,

Thank you for your message! We're delighted to hear that you have received your debit card.

As a reminder, your ticket has been successfully resolved:

- Ticket ID: INC000000001
- Status: Resolved
- Issue: Debit card replacement has not been delivered after 10 business days.
- Resolution: Replacement card has been delivered.

If you have any further questions or need assistance in the future, please don't hesitate to reach out. We appreciate your trust in us and look forward to serving you again.

Thank you for banking with us!

Was this helpful?



Type your message...



Figure 9: Output Screenshot | Intent - Positive | Resolution Confirmation

Intent – Inquiry

- **Ticket Inquiry by providing ticket id in the dedicated UI field**

Banking Customer Support AI Assistant

Multi-Agent Customer Support System using LangGraph

Customer Assistant

Response Evaluation (LLM-as-Judge)

Current Customer Profiling

Agent Execution Trace

Intent Classification Evaluation (Manual Test-Cases)

Customer Details

Customer Name

John Smith

Customer Email

john.smith@gmail.com

Ticket Number

INC0000000001

Inquiry

New Session

Chat History

Welcome John Smith.

I found your existing ticket details below:

Field	Value
Ticket ID	INC0000000001
Status	Resolved
Issue Description	Debit card replacement has not been delivered after 10 business days.
Tech Notes	Replacement card dispatch verification pending.
Created	2026-06-01 09:15:00
Last Updated	2026-06-23 16:50:02

How may I assist you today?

Please provide status of the ticket

Hello John Smith,

Thank you for your inquiry regarding the status of your ticket. Here are the details:

- Ticket ID: INC0000000001
- Current Status: Resolved
- Issue Description: Debit card replacement has not been delivered after 10 business days.
- Last Updated: June 23, 2026

It appears that your issue has been marked as resolved. However, if you are still experiencing any delays or have further questions, please let us know, and we will be happy to assist you further!

Thank you for your patience!

Was this helpful?

Type your message...

Figure 10: Output Screenshot | Intent – Inquiry | Ticket ID in Field

- **Ticket Inquiry by not providing any ticket (neither in the query section nor in the dedicated field)**

Banking Customer Support AI Assistant

Multi-Agent Customer Support System using LangGraph

Customer Assistant

Response Evaluation (LLM-as-Judge)

Current Customer Profiling

Agent Execution Trace

Intent Classification Evaluation (Manual Test-Cases)

Customer Details

Customer Name

John Smith

Customer Email

john.smith@gmail.com

Ticket Number

Inquiry

New Session

Chat History

Welcome John Smith.

How may I assist you today?

Please provide the the status of the ticket

Hello John Smith,

Thank you for reaching out. Unfortunately, I could not identify a valid ticket number in your request. To assist you further, please provide a valid Ticket ID. The format should be: INC + 10 digits (for example: INC0000000001).

Once I have the correct Ticket ID, I'll be able to provide you with the status of your inquiry.

Thank you for your understanding!

Was this helpful?

Type your message...

Figure 11: Output Screenshot | Intent – Inquiry | No Ticket ID Provided

– Ticket Inquiry by providing ticket id directly in the query



Banking Customer Support AI Assistant

Multi-Agent Customer Support System using LangGraph

Customer Assistant Response Evaluation (LLM-as-Judge) Current Customer Profiling Agent Execution Trace Intent Classification Evaluation (Manual Test-Cases)

Customer Details

Customer Name

John Smith

Customer Email

john.smith@gmail.com

Ticket Number

INC0000000002

Inquiry

New Session

Chat History



Welcome John Smith.

How may I assist you today?



Please provide the status of INC0000000002



Hello John Smith,

Thank you for your inquiry regarding the status of your ticket, INC0000000002. Here are the details:

- **Ticket ID:** INC0000000002
- **Current Status:** Resolved
- **Issue Description:** Unable to login to internet banking account despite entering correct credentials.
- **Technical Comments:** Password reset completed and you confirmed successful login.
- **Last Updated:** June 3, 2026, at 15:45

We appreciate your patience throughout this process. If you have any further questions or need additional assistance, please feel free to reach out.

Best regards!

Was this helpful?



Type your message...



Figure 12: Output Screenshot | Intent – Inquiry | Ticket ID in User Query

Intent – Negative

– Raise a complaint to create a new ticket



Banking Customer Support AI Assistant

Multi-Agent Customer Support System using LangGraph

Customer Assistant Response Evaluation (LLM-as-Judge) Current Customer Profiling Agent Execution Trace Intent Classification Evaluation (Manual Test-Cases)

Customer Details

Customer Name

John Smith

Customer Email

john.smith@gmail.com

Ticket Number

INC0000000021

Negative / Complaint

New Session

Chat History



Welcome John Smith.

How may I assist you today?



I am not able to login to Bank Customer Portal. Please assist in resetting the password.



Hello John Smith,

I understand how frustrating it can be to experience login issues with the Bank Customer Portal. A new support ticket has been created to assist you with resetting your password. Here are the details:

- **Ticket ID:** INC0000000021
- **Status:** Open
- **Issue Summary:** Unable to login to Bank Customer Portal and requesting assistance in resetting the password.

Our support team is currently reviewing your request and will take the necessary steps to assist you. You can expect an update soon regarding the next steps.

If you have any further questions or concerns in the meantime, please feel free to reach out.

Was this helpful?



Type your message...



Figure 13: Output Screenshot | Intent – Negative | Complaint for a New Ticket

- Upon Inquiry, a ticket is found as Resolved. Negative Feedback on follow-up creates a new ticket for a follow-on investigation

Banking Customer Support AI Assistant

Multi-Agent Customer Support System using LangGraph

Customer Assistant Response Evaluation (LLM-as-Judge) Current Customer Profiling Agent Execution Trace Intent Classification Evaluation (Manual Test-Cases)

Customer Details

Customer Name
John Smith

Customer Email
john.smith@gmail.com

Ticket Number
INC0000000022

Negative / Complaint

New Session

Chat History

Welcome John Smith.

I found your existing ticket details below:

Field	Value
Ticket ID	INC0000000001
Status	Resolved
Issue Description	Debit card replacement has not been delivered after 10 business days.
Tech Notes	Replacement card dispatch verification pending.
Created	2026-06-01 09:15:00
Last Updated	2026-06-23 16:50:02

How may I assist you today?

The debit card is still not delivered to me as yet. What is taking so long? How did the status change to Resolved when I still have not received the card?

Dear John Smith,

I sincerely apologize for the inconvenience you are experiencing with the delivery of your debit card. I understand how frustrating this situation must be for you, especially since the previous ticket was marked as Resolved.

To clarify, your previous ticket (INC0000000001) regarding the delay in receiving your debit card has indeed been marked as Resolved. However, since you have reported that the issue persists, I have created a new ticket for further investigation. Here are the details:

- New Ticket ID:** INC0000000022
- Status:** Open
- Issue Summary:** Follow-on investigation for ticket (INC0000000001) regarding the undelivered debit card.

Our support team is actively looking into this matter, and I assure you that we will keep you updated on any progress. Thank you for your patience as we work to resolve this issue.

If you have any further questions or concerns, please feel free to reach out.

Best regards,

Was this helpful? 👍 👎

Figure 14: Output Screenshot / Intent – Negative / Complaint for an Existing Resolved Ticket

- Upon Inquiry, a ticket is found as Open. Negative Feedback on follow-up doesn't create a new ticket to avoid duplication

Banking Customer Support AI Assistant

Multi-Agent Customer Support System using LangGraph

Customer Assistant Response Evaluation (LLM-as-Judge) Current Customer Profiling Agent Execution Trace Intent Classification Evaluation (Manual Test-Cases)

Customer Details

Customer Name
John Smith

Customer Email
john.smith@gmail.com

Ticket Number
INC0000000021

Negative / Complaint

New Session

Chat History

Welcome John Smith.

I found your existing ticket details below:

Field	Value
Ticket ID	INC0000000021
Status	Open
Issue Description	Unable to login to Bank Customer Portal and requesting assistance in resetting the password.
Tech Notes	None
Created	2026-06-23 17:17:15
Last Updated	2026-06-23 17:17:15

How may I assist you today?

I cannot login to portal

Hello John Smith,

I understand how frustrating it can be to have trouble logging into your account, and I appreciate your patience as we work to resolve this issue.

Currently, there is an active support ticket for your login problem. Here are the details:

- Ticket ID:** INC0000000021
- Current Status:** Open
- Issue Summary:** Unable to login to Bank Customer Portal and requesting assistance in resetting the password.
- Latest Update:** No updates available yet.
- Next Action:** The support team is actively working on your issue.

Please rest assured that we are committed to resolving your login problem as quickly as possible. If you have any further questions or additional information to provide, feel free to let me know!

Thank you for your feedback.

Type your message...

Figure 15: Output Screenshot / Intent – Negative / Complaint for an Existing Open Ticket

Section 7 – Key Capabilities & Innovation Highlights

- The Banking Customer Support AI Assistant combines Agentic AI, workflow orchestration, customer intelligence and automated quality assurance to deliver a customer support experience that is accurate, personalized, and operationally governed.

Key Capabilities

1. Multi-Agent Workflow Architecture

- The solution utilizes a LangGraph-based multi-agent architecture where specialized agents perform distinct responsibilities such as intent classification, ticket management, customer intelligence generation, response generation, and response evaluation.
- **Business Value**
 - ✓ Improves modularity and maintainability.
 - ✓ Enables separation of responsibilities.
 - ✓ Supports scalable workflow orchestration.

2. Intelligent Query Understanding

- The Query Classification Agent analyses customer requests to determine intent, interaction type, ticket references, and resolution confirmation status before workflow execution begins.
- **Business Value**
 - ✓ Improves workflow routing accuracy.
 - ✓ Reduces incorrect ticket handling.
 - ✓ Enhances customer experience through context-aware processing.

3. Business Workflow Automation

- The solution automates ticket lifecycle decisions by intelligently determining whether existing tickets should be reused, resolved, or whether new tickets should be created.
- **Business Value**
 - ✓ Reduces manual decision making.
 - ✓ Ensures workflow consistency.
 - ✓ Aligns responses with business process rules.

4. Customer Intelligence Framework

- The solution generates customer-specific intelligence using historical interactions, customer feedback and evaluation results.
- The generated intelligence is used to personalize future customer interactions while maintaining a consistent customer experience.
- **Business Value**
 - ✓ Enables personalized customer engagement.
 - ✓ Improves communication effectiveness.
 - ✓ Supports customer-centric service delivery.

5. LLM-as-Judge Quality Assurance

- Every AI-generated response is evaluated before delivery using an automated quality assessment framework.
- Responses are assessed against predefined quality dimensions to ensure accuracy, compliance and customer experience standards.
- **Business Value**
 - ✓ Improves response quality.
 - ✓ Reduces hallucinations and inaccuracies.
 - ✓ Increases trust in AI-generated responses.

6. Reflection-Based Self-Correction

- The solution incorporates an automated reflection mechanism that uses evaluation feedback to improve responses when quality standards are not met.
- **Business Value**
 - ✓ Creates a self-improving workflow.
 - ✓ Enhances response quality.
 - ✓ Reduces manual intervention.

7. End-to-End Traceability

- The Execution traces are captured throughout the workflow lifecycle, providing visibility into decisions, routing actions, ticket operations, and evaluation outcomes.
- **Business Value**
 - ✓ Improves transparency.
 - ✓ Supports troubleshooting and debugging.
 - ✓ Enhances governance and auditability.

8. Integrated Evaluation, Analytics & Monitoring

- The solution combines response evaluation, customer intelligence analytics, and workflow monitoring into a unified governance framework.
- **Business Value**
 - ✓ Enables operational visibility.
 - ✓ Supports continuous improvement.
 - ✓ Facilitates AI governance and oversight.

Innovation Highlights

- Agentic AI workflow orchestration using LangGraph.
- Customer intelligence-driven response personalization.
- Automated ticket lifecycle management.
- LLM-as-Judge response evaluation framework.
- Reflection-based response optimization.
- End-to-end workflow traceability.
- Integrated analytics and governance capabilities.
- Stateful conversational support with persistent memory.

Section 8 – Challenges Faced & Future Enhancements

Challenges Faced

1. **Ticket Lifecycle Management**
 - Handling complex ticket workflows required careful business rule design.
 - Special consideration was needed for scenarios where customers reported that previously resolved issues were still occurring.
 - The solution introduced dedicated workflow handling to ensure that resolved tickets remained unchanged while follow-on tickets were created for continued investigation.
2. **Customer Intelligence Calibration**
 - Generating meaningful customer intelligence without over-interpreting limited historical data presented a significant challenge.
 - Several iterations were required to ensure that customer profiles remained evidence-based, explainable and actionable.
3. **Response Quality Evaluation**
 - Defining effective evaluation criteria and balancing strictness versus practicality required multiple iterations.
 - The evaluation framework needed to accurately identify genuine response deficiencies while avoiding unnecessary regeneration cycles.
4. **Prompt Engineering Complexity**
 - The solution relies on multiple specialized prompts across different workflow stages.
 - Ensuring consistency between agents while maintaining clear responsibilities required careful prompt design and testing.

Future Enhancements

1. **Knowledge Base Integration**
 - Integrate enterprise knowledge repositories to provide richer contextual responses and improve issue resolution capabilities.
2. **Advanced Customer Sentiment Analysis**
 - Incorporate deeper sentiment analysis to better understand customer emotions and adapt communication strategies dynamically.
3. **Human-in-the-Loop Escalation**
 - Introduce escalation workflows that allow human agents to review and intervene in high-risk or complex customer interactions.
4. **Enhanced Customer Intelligence**
 - Expand customer profiling capabilities through additional behavioural, interaction and service quality insights.
5. **Advanced Analytics & Reporting**
 - Introduce trend analysis, operational dashboards and predictive insights to support service management and governance activities.
6. **AIOps & Proactive Support**
 - Leverage predictive analytics to identify recurring issues, anticipate customer concerns and proactively recommend support actions.